CASE STUDY

RESEARCH & TECHNOLOGY ORGANIZATION

A leading UK Clinical Research Organization (CRO) faced operational inefficiencies due to fragmented and disjointed methods for recording data across multiple laboratories.

These inconsistent approaches led to operational inefficiencies, increased maintenance costs, limited collaboration, and compromised data integrity.

To address these challenges, the CRO adopted **Logilab ELN,** a cloud-based electronic lab notebook designed to drive digital transformation.

By centralizing data and enabling seamless collaboration across laboratories, the CRO has significantly improved efficiency and overall project coordination. This transformation has fostered a unified, agile research environment that accelerates innovation.

WHAT'S UP WITH CROSS-DISCIPLINARY RESEARCH LABS

A leading independent Clinical Research Organization (CRO) in the United Kingdom supports companies in developing new products and processes, guiding them from early concept development through to research and development, manufacturing, and commercialisation.

With specialist teams spanning biologics, biotechnology, advanced materials, electronics, and pharmaceuticals, the CRO plays a vital role in bridging the gap between early-stage research and full-scale production. Its mission is to accelerate innovation, reduce risk, and support both startups and large enterprises in creating innovative solutions.

Despite their critical role in advancing innovation across diverse sectors, the organization faced operational challenges due to the use of multiple, disconnected systems for capturing and storing data across laboratories. Given the scope and impact of their work, it was essential to implement an efficient, centralized system to streamline data capture and lab data management.

Their challenges included,

Inconsistent Process: There was no standardized method for recording and storing data across the organization's laboratories. While some relied on paper records, others used Excel spreadsheets or isolated digital tools, leading to inconsistent formats, incomplete metadata, and challenges in maintaining data integrity.

Disjointed Systems: Each laboratory used different digital solutions to capture data, leading to siloed information and a lack of integration across platforms. This incohesive approach made it difficult to transfer knowledge between teams, slowing down collaboration and decision-making.

Expensive Maintenance: Maintaining multiple, isolated systems across laboratories required extensive IT resources and incurred high costs. Each platform involved separate licensing fees, updates, and support services, creating a financial burden.

Fragmented Expertise: Since each laboratory relied on unique tools and processes, specific subject matter experts were needed to manage and operate these systems. This specialization limited flexibility and made it difficult for staff to move between projects or teams without retraining.

Data Silos: The use of isolated systems meant that data transfer between teams was complex. Sharing experimental data and related context across laboratories was often cumbersome and error-prone, leading to delays in project timelines and reduced overall effectiveness.

Limited Collaboration: Without a centralized platform, teams across different labs struggled to collaborate effectively. Communication gaps and incompatible data formats hindered real-time data sharing and collaborative research.

WHY LOGILAB ELN IS BETTER THAN PAPER OR EXCEL-BASED SYSTEMS

Logilab ELN provides a centralized, cloud-based platform that streamlines data management with real-time access and automated processes. Moving beyond traditional paper and spreadsheet methods, it minimizes errors, fosters collaboration across locations, and ensures adherence to regulatory compliance requirements. Designed for today's dynamic labs, the system boosts lab productivity, strengthens data integrity, and scales effortlessly to support growing research needs.

GET THE PICTURE

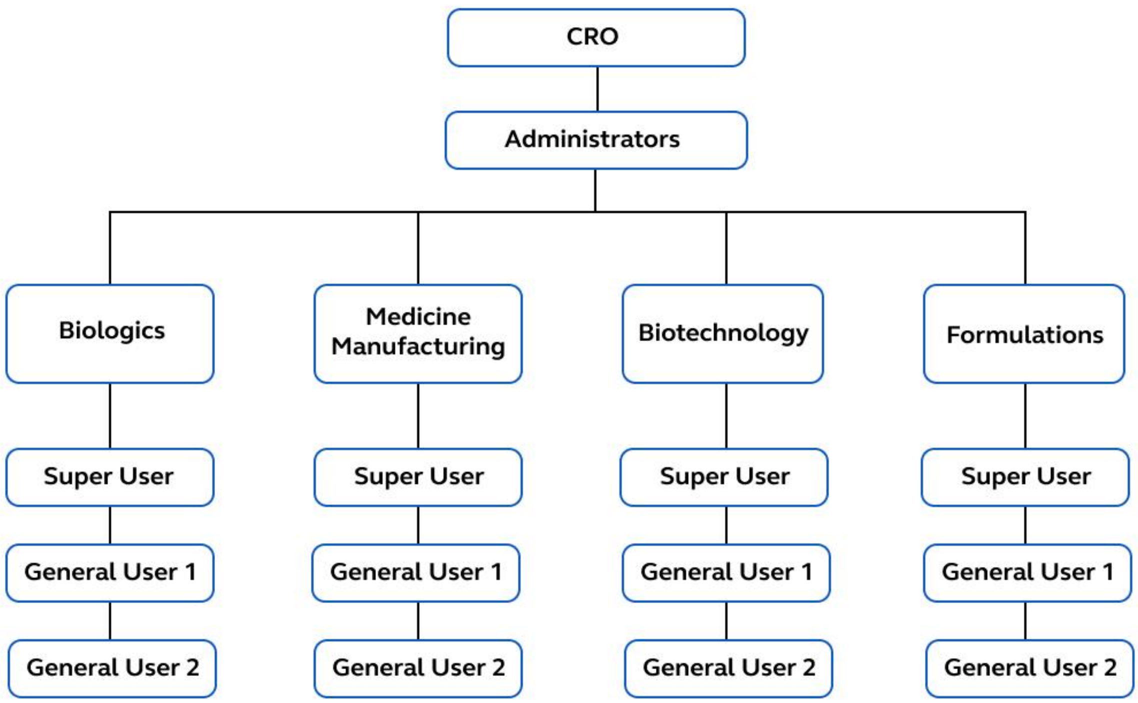
With multiple labs and teams working in different technical areas, the CRO generates large amounts of experimental data every day. However, each group had their own method of recording and storing data; some used paper-based documentation, while others relied on spreadsheets or standalone digital tools. This lack of standardization led to data fragmentation, inconsistent formats, and incomplete metadata. Maintaining these disjointed systems became increasingly expensive, and as experimental projects grew in volume and intricacy, the need for a scalable solution became urgent.

The CRO also required a system that could support structured project management and enable effective collaboration across multiple teams and locations. Their research activities involved multi-user experiments that needed to be tracked, reviewed, and approved within controlled workflows. The lack of a centralized platform made coordination difficult, especially when managing permissions, timelines, and shared resources, which led to roadblocks and project delays. To address this, the RTO sought a configurable solution that allowed teams to plan, record, and monitor experimental work in a connected environment.

Agaram Technologies proposed the implementation of **Logilab ELN**, an electronic lab notebook designed to digitize laboratories and streamline lab data management, to resolve these issues. The platform's multi-site architecture allows for the seamless operation of different business units under one unified system, while maintaining logical separation of data and access. This ensures both data security and operational flexibility.

UNIFYING RESEARCH LABS & ELEVATING INNOVATION

The implementation was rolled out in strategic phases which included laboratories focused on Biologics, Biotechnology, Medicines Manufacturing and Formulations. With over 300 licensed users across multiple laboratories, Logilab ELN supports a broad range of CRO's research activities.



Multi-Site Collaboration

OPTIMIZED LAB DATA MANAGEMENT

Logilab ELN significantly streamlined data management by providing a clear and organized way to handle samples and materials across laboratories. The sample management feature in the inventory allowed users to track the entire lifecycle of a sample, including its origin, experimental use, and final disposition. The ability to trace sample lineage supported both regulatory compliance and scientific reproducibility.

The inventory also provided biologics and biotechnology teams with an efficient way to manage cell banks and other biologically sensitive materials. It offered the flexibility to define custom parameters and metadata fields, ensuring each sample could be tracked according to its specific attributes. Overall, Logilab ELN improved the accuracy, consistency, and accessibility of data, making it easier for teams to record, store and manage their data efficiently.

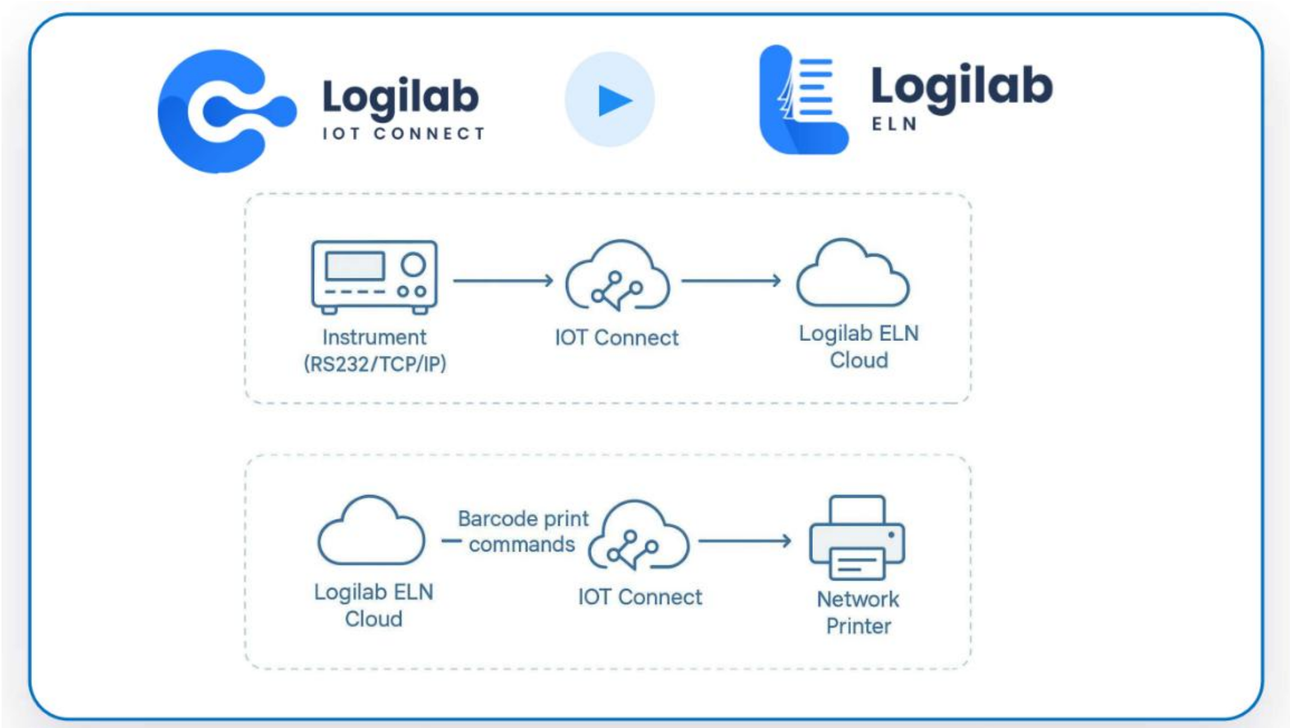
ENHANCED PROJECT MANAGEMENT

The implementation of Logilab ELN overhauled project management within the RTO by creating a unified platform for data capture and collaboration. Logilab ELN enabled smooth, real-time sharing of experimental data among teams, eliminating communication barriers and breaking down data silos. This facilitated more efficient cross-disciplinary project management, allowing different teams to coordinate seamlessly in a centralized platform. Automated workflows and standardized templates ensured consistent processes across projects, reducing errors and easing staff transitions between teams. The integrated audit trails and version control also increased transparency and accountability. As a result, projects were better aligned with timelines, enhancing overall productivity and innovation.

CONTROLLED DATA ACCESS & SITE CONFIGURATIONS

The CRO required the ability to manage projects without compromising confidentiality or compliance. Logilab ELN offers robust role-based access control, allowing administrators to configure user permissions based on project, laboratories, or external stakeholder needs. The ability to create dedicated sites for specific projects or collaborations within the ELN ensures that sensitive information is kept secure, accessible only by authorized users.

CENTRALIZED LABELING VIA loT INTEGRATION



Label Printing via Logilab IoT Workflow

Mock Label Example

The previous label printing process was cumbersome, requiring users to manually connect to individual printers, which often caused delays and confusion. To overcome this, Agaram introduced Logilab loT

Connect, enabling remote queuing of print jobs to designated printers across the network. This centralized system improved print job tracking and simplified the label printing process by enabling easier monitoring and management. It also ensured consistent label quality across all laboratories, supporting smoother operations throughout the CRO.

KEY BENEFITS

- **Enhancing Data Capture through Digitization:** The automation of experimental data capture and inventory tracking significantly reduced the time spent on manual documentation. With configurable templates and dynamic data fields, users could create, retrieve, and manage experimental records faster and more reliably across laboratories.
- **Cloud-Based Centralized Control:** Logilab ELN provided a centralized, cloud-based environment for managing experiments, inventory, and collaboration across multiple laboratories. It provided real-time visibility and control over data, enabling faster decision-making and project coordination across locations, eliminating geographical barriers.
- **Accessibility:** With Single Sign-On (SSO) integration, users can securely access Logilab ELN using their credentials from anywhere, enabling location-independent access to lab data.
- **Internal and External Collaboration:** The platform's multi-site capability enabled smooth collaboration between laboratories and with external stakeholders. Projects could be logically separated while still allowing controlled cross-site access, aiding interdisciplinary research.
- **Enhanced Data Integrity and Compliance:** Logilab ELN ensures adherence to 21 CFR Part 11, GxP and ALCOA+ principles, helping maintain data accuracy, reliability, and traceability to support regulatory compliance
- **Audit Readiness:** All data operations within the ELN are automatically logged and timestamped, supporting audit trails for both internal reviews and regulatory inspections.
- **Optimized Resource Utilization:** By eliminating the need for multiple siloed systems, the CRO was able to reduce software overhead and IT maintenance efforts. Teams could now access shared resources and infrastructure, improving operational efficiency.

By creating a more connected, efficient, and innovative work environment across multiple laboratories, Logilab ELN has significantly improved the CRO's research operations.

To see Logilab ELN in action, request a demo and get a detailed overview here.

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